

NeoPixels

DO

OUTPUT

LIGHT

What does it do?

A NeoPixel is a full-colour LED with a tiny controller built into it.

One data wire carries colour instructions, and each pixel passes the remaining instructions onwards.

The 3.3 V-compatible JST breakout shown here can connect directly to a Pi or Pico.

Uses



Gaming PC lighting

DETECT

Read lighting selection



DECIDE

Which colour should each pixel be?



DO

Light individual pixels



Disco shoelace meter

DETECT

Read shoe movement



DECIDE

Which light pattern should run?



DO

Animate the pixels



Pixel
RGB LED and controller

IN
Data from the board

OUT
Data to another pixel

VDD and GND
3.3 V power

Code

● Raspberry Pi

```
import board
import neopixel

pixels = neopixel.NeoPixel(board.D18, 1)

pixels[0] = (255, 0, 80)
```

● Raspberry Pi Pico

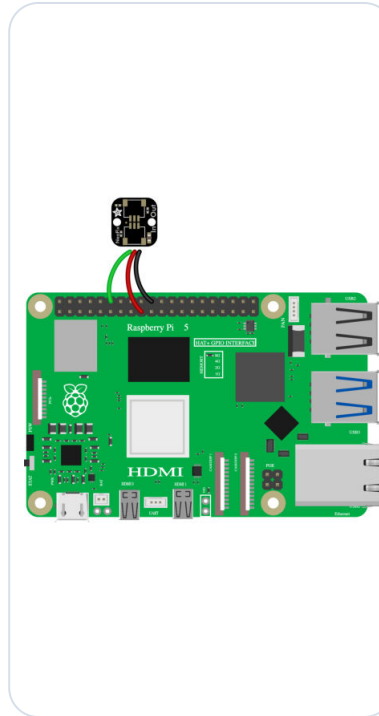
```
from machine import Pin
from neopixel import NeoPixel

pixels = NeoPixel(Pin(28), 1)
pixels[0] = (255, 0, 80)
pixels.write()
```

This wiring is for the pictured 3.3 V-compatible breakout. Generic 5 V strips usually need a separate supply and a logic-level shifter.

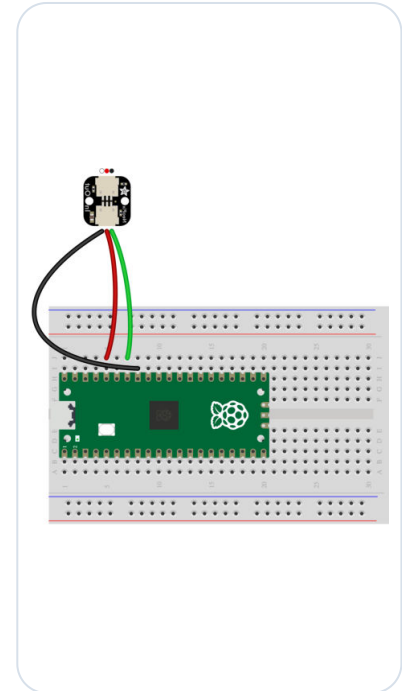
Wiring

RASPBERRY PI 5



GPIO18 → IN, **3V3** → VDD, **GND** → GND

PICO



GP28 → IN, **3V3 OUT** → VDD, **GND** → GND